

# More Interesting Examples of Level Curves

Calculus and Ordinary Differential Equations

Supplementary tutorial examples

For a function

$$f : \mathbb{R}^2 \rightarrow \mathbb{R},$$

the **level curve** or **contour corresponding to the value  $k$**  is

$$f(x, y) = k.$$

All points on the same contour therefore have exactly the same function value.

In applications,  $f(x, y)$  might represent temperature, height, pressure, potential energy, stress, concentration, or some other physical quantity.

## 1 Elliptical contours: different behaviour in different directions

Consider

$$f(x, y) = 3x^2 + 2y^2.$$

The level curve corresponding to  $k > 0$  is

$$3x^2 + 2y^2 = k.$$

Dividing by  $k$ ,

$$\frac{x^2}{k/3} + \frac{y^2}{k/2} = 1.$$

Hence each level curve is an ellipse centred at the origin, with semi-axes

$$a_x = \sqrt{\frac{k}{3}}, \quad a_y = \sqrt{\frac{k}{2}}.$$

Since

$$\sqrt{\frac{k}{3}} < \sqrt{\frac{k}{2}},$$

the ellipse is narrower in the  $x$ -direction than in the  $y$ -direction.

### Point to emphasise in class

Why are the ellipses narrower along  $x$ ?

If we move along the  $x$ -axis,

$$f(x, 0) = 3x^2,$$

whereas along the  $y$ -axis,

$$f(0, y) = 2y^2.$$

Thus the same change in distance produces a larger increase in  $f$  along the  $x$ -direction. A contour of fixed value  $k$  is therefore reached sooner when moving along  $x$ .

### Engineering interpretation / application

Suppose  $f(x, y)$  represents a simplified **potential energy** or **temperature increase** near the centre of a component.

Elliptical contours indicate that the physical response is *direction-dependent*. For instance, heat might spread more easily in one direction than another because the material conducts heat differently along different axes.

This is realistic for materials that are **anisotropic**, such as composites, layered materials, certain crystals, or fibre-reinforced structures.

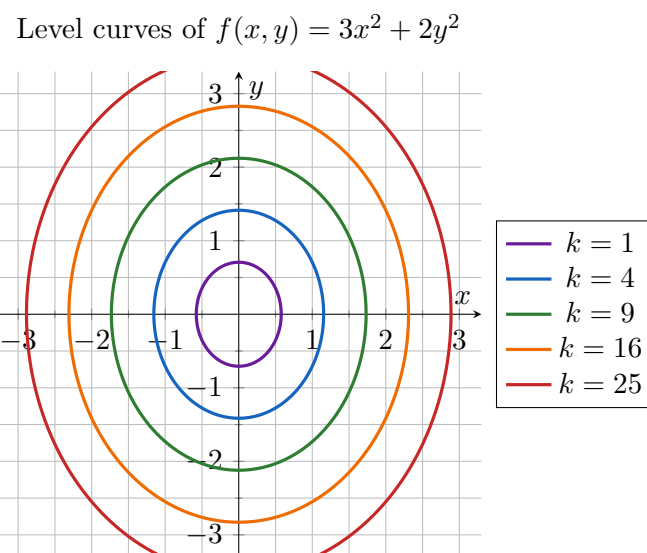


Figure 1: Elliptical contours. The function grows more rapidly in the  $x$ -direction than in the  $y$ -direction.

## 2 Rotated elliptical contours: a cross-term changes the preferred directions

Now consider

$$f(x, y) = x^2 + xy + y^2.$$

The level curves satisfy

$$x^2 + xy + y^2 = k.$$

This is again an ellipse for every  $k > 0$ , but the cross-term  $xy$  means that its principal axes are **not aligned with the  $x$ - and  $y$ -axes**.

To see the orientation more clearly, introduce rotated coordinates

$$u = \frac{x+y}{\sqrt{2}}, \quad v = \frac{x-y}{\sqrt{2}}.$$

Then

$$x = \frac{u+v}{\sqrt{2}}, \quad y = \frac{u-v}{\sqrt{2}},$$

and substitution gives

$$x^2 + xy + y^2 = \frac{3}{2}u^2 + \frac{1}{2}v^2.$$

Thus

$$\frac{3}{2}u^2 + \frac{1}{2}v^2 = k$$

is an ordinary ellipse in the rotated  $(u, v)$ -coordinate system.

Its axes correspond to the diagonal directions

$$y = x \quad \text{and} \quad y = -x.$$

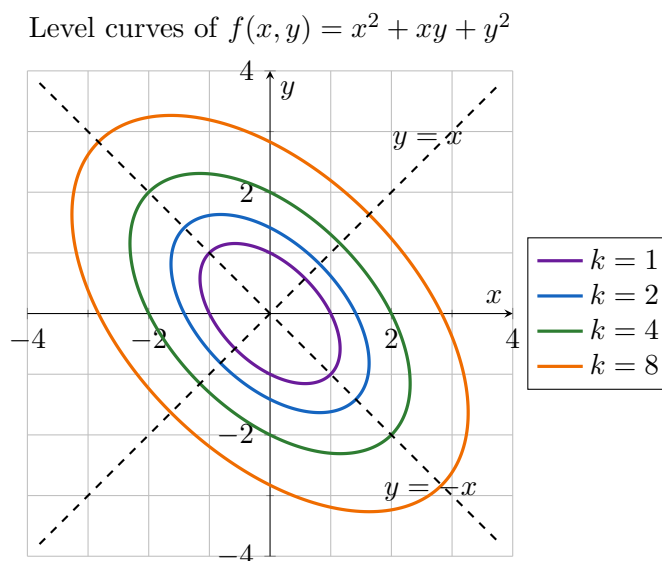


Figure 2: Rotated elliptical contours caused by the cross-term  $xy$ .

### Point to emphasise in class

A useful general observation is that a quadratic expression

$$Ax^2 + Bxy + Cy^2$$

need not have its natural directions aligned with the coordinate axes.

The  $xy$ -term is a signal that the geometry may be **rotated**. In more advanced mathematics this is handled systematically using matrices, eigenvalues and eigenvectors.

### Engineering interpretation / application

Suppose  $f(x, y)$  represents **elastic strain energy**, deformation, or stress in a thin sheet. If the contours are rotated, the directions in which the material is easiest or hardest to deform need not coincide with the horizontal and vertical axes.

This occurs naturally in engineering when:

- fibres in a composite are placed diagonally;
- a material has an internal crystal direction;
- a component is loaded at an angle;
- stresses interact in more than one coordinate direction.

Thus the orientation of the contours contains physical information, not merely geometrical decoration.

### 3 Shifted circular contours: locating a source

Consider

$$f(x, y) = (x - 2)^2 + (y + 1)^2.$$

The level curves satisfy

$$(x - 2)^2 + (y + 1)^2 = k.$$

Therefore each contour is a circle with centre

$$(2, -1)$$

and radius

$$\sqrt{k}.$$

Unlike the familiar example

$$x^2 + y^2 = k,$$

these circles are not centred at the origin.

Level curves of  $f(x, y) = (x - 2)^2 + (y + 1)^2$

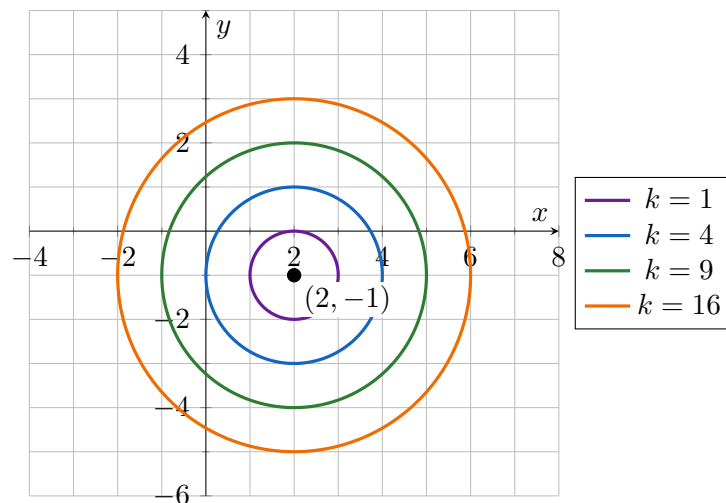


Figure 3: Concentric contours centred at a point other than the origin.

### Engineering interpretation / application

Imagine that  $(2, -1)$  is the position of a small electronic component that acts as a **heat source** on a circuit board.

A physically more realistic temperature function might decrease with distance from the source, for example

$$T(x, y) = T_{\text{ambient}} + Ae^{-\alpha((x-2)^2+(y+1)^2)},$$

where  $A > 0$  and  $\alpha > 0$ .

A temperature contour

$$T(x, y) = T_0$$

can be rearranged into

$$(x - 2)^2 + (y + 1)^2 = \text{constant}.$$

Hence the equal-temperature curves are again circles centred on the heat source.

An engineer can use such a contour plot to identify:

- where the board may exceed a safe operating temperature;
- where a heat sink should be placed;
- how far a temperature-sensitive component should be kept from the source;
- whether additional cooling is necessary.

## 4 A non-conic example: contours with four-fold symmetry

For a less familiar example, consider

$$f(x, y) = x^4 + y^4.$$

The level curves satisfy

$$x^4 + y^4 = k, \quad k > 0.$$

Equivalently,

$$\left(\frac{x}{k^{1/4}}\right)^4 + \left(\frac{y}{k^{1/4}}\right)^4 = 1.$$

These are examples of **superellipses**. They resemble rounded squares rather than ordinary ellipses.

Level curves of  $f(x, y) = x^4 + y^4$

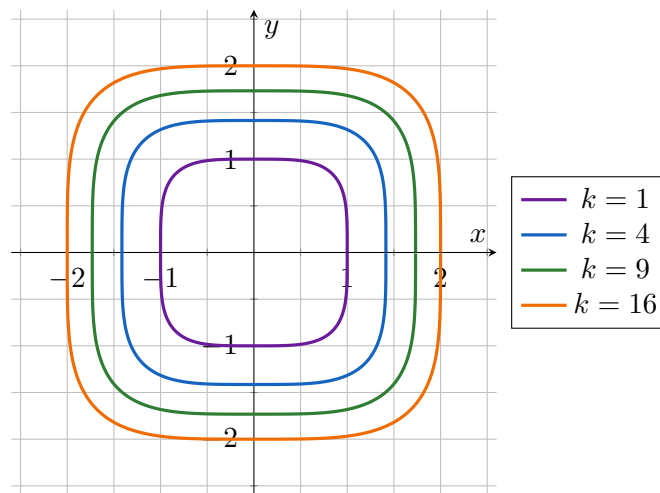


Figure 4: Contours of  $x^4 + y^4$ , which resemble rounded squares rather than conic sections.

### Point to emphasise in class

This example is useful because students may unconsciously assume that level curves must always be circles, ellipses, parabolas, or hyperbolas.

There is no such requirement.

A contour is simply the solution set of

$$f(x, y) = k.$$

Depending on  $f$ , it may have almost any shape.

### Engineering interpretation / application

Functions involving fourth powers can arise as simplified **energy or penalty functions**.

For example, suppose deviations in the  $x$ - and  $y$ -directions become disproportionately costly when they become large:

$$E(x, y) = x^4 + y^4.$$

The contours show combinations of  $x$  and  $y$  that produce the same energy or penalty.

Similar ideas occur in optimisation, control systems, structural modelling, and numerical design, where a designer may wish to visualise regions of equal cost or equal performance.

## 5 A useful comparison for students

| Function                          | Contour equation            | Shape / interpretation                                                               |
|-----------------------------------|-----------------------------|--------------------------------------------------------------------------------------|
| $f(x, y) = x^2 + y^2$             | $x^2 + y^2 = k$             | Circles centred at the origin. Isotropic behaviour: all directions are equivalent.   |
| $f(x, y) = 3x^2 + 2y^2$           | $3x^2 + 2y^2 = k$           | Ellipses. The function changes at different rates in the two coordinate directions.  |
| $f(x, y) = x^2 + xy + y^2$        | $x^2 + xy + y^2 = k$        | Rotated ellipses. The natural directions are not the coordinate axes.                |
| $f(x, y) = (x - 2)^2 + (y + 1)^2$ | $(x - 2)^2 + (y + 1)^2 = k$ | Circles centred at $(2, -1)$ . Useful for modelling a source away from the origin.   |
| $f(x, y) = x^4 + y^4$             | $x^4 + y^4 = k$             | Rounded-square or superelliptic contours; an example beyond ordinary conic sections. |

### Central interpretation.

If two points  $P_1$  and  $P_2$  lie on the same contour, then

$$f(P_1) = f(P_2).$$

Hence, depending on the application, the two locations have the same temperature, pressure, height, potential energy, concentration, stress level, or other modelled quantity.

What changes from one example to another is the *geometry of all points having the same value*.